

-- HDOC Day-5 Assignment --

- Question 1 :

Sqaure plot side length

1 – input, output and formula identification

- Input ; perimeter of the sqaure plot (p) in meters.
- Output ; side length of the sqaure plot (s) in meters.
- Formula : $\text{side} = \text{perimeter}/4$

2 – test case table

Test case	Input (p)	Expected output (s)	Actual output	Result (pass/fail)
1.	20	5.00	5.00	pass
2.	30	7.50	7.50	pass
3.	35	8.75	8.75	pass

3 – algorithm ;

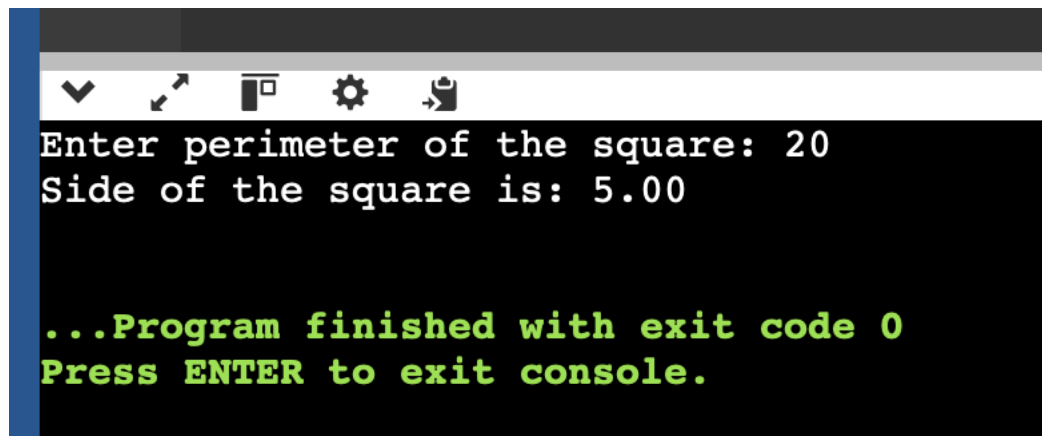
- Start the program.
- Declaring variable “perimeter” and “side” .
- Ask the user to enter the perimeter of the sqaure plot in meters.
- Reading and store the user input in “perimeter” variable.
- Calculating the side using the formula.
- Displaying the calculated side length formatted to 2 decimal places.
- Ending the program.

4 – c code ;

```
#include <stdio.h>
```

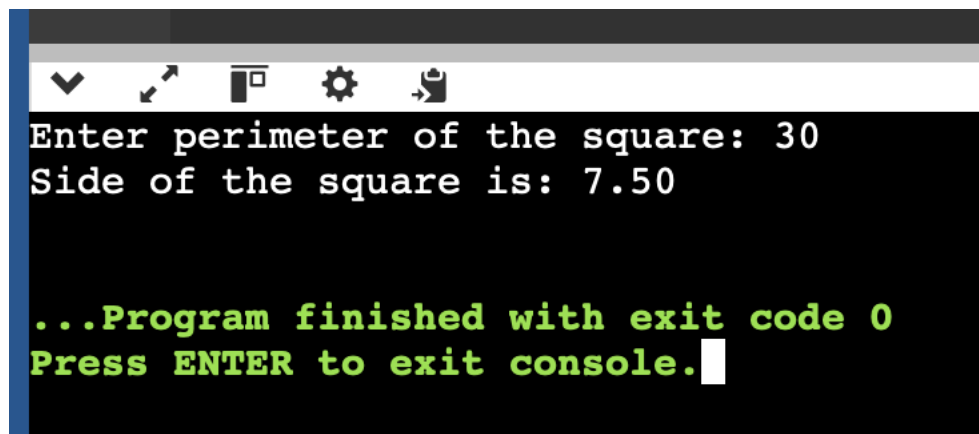
```
int main() {  
    float p, side;  
  
    // taking perimeter input from user  
    printf("Enter perimeter of the square: ");  
    scanf("%f", &p);  
  
    // formula to calculate side  
    side = p / 4;  
  
    // printing result  
    printf("Side of the square is: %.2f\n", side);  
  
    return 0;  
}
```

5- execution results :



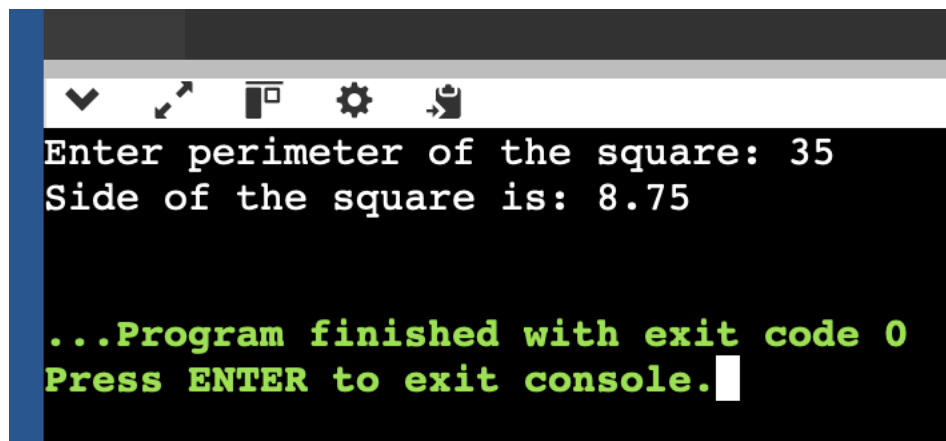
```
Enter perimeter of the square: 20
Side of the square is: 5.00

...Program finished with exit code 0
Press ENTER to exit console.
```



```
Enter perimeter of the square: 30
Side of the square is: 7.50

...Program finished with exit code 0
Press ENTER to exit console.
```



```
Enter perimeter of the square: 35
Side of the square is: 8.75

...Program finished with exit code 0
Press ENTER to exit console.
```

- Question 2:

Rhombus area , perimeter and coloring costs

1 – input , output, and formula identification

- Input :

- Side length of the rhombus (s) in cms.

- Height of the rhombus (h) in cms.
- Coloring rate -/ square centimeter (r).
- Output :
 - Area of the rhombus in square. Cm .
 - Perimeter of the rhombus in cm
 - Total coloring cost.
- Formula ;
 - $\text{Area} = s * h$
 - $\text{Perimeter} = 4 * s$
 - $\text{Coloring cost} = \text{area} * r$

2 – test case table :

Test case	Input (s,h,r)	Expected op (area, p , cost)	Actual output	Result (pass/fail)
1.	5,10,4	Area = 50.00 P = 20.00 Cost = 200.00	Area=50.00 P=20.00 Cost=200.00	pass
2.	10,12,4	Area = 120.00 P = 40.00 Cost=480.00	Area=120.00 P=40.00 Cost=480.00	pass
3.	15,20,4	Area=300.00 P=60.00 Cost=1200.00	Area=300.00, P=60.00 Cost=1200.00	pass

3 – algorithm :

- Start the program
- Declare variables for side, height, rate , area, perimeter and cost.
- Ask the user to enter side length, height, and coloring rate per sq unit.
- Read and store the values given by user using scanf.

- Compute the area using formula ($\text{area} = \text{side} * \text{height}$).
- Compute the perimeter using ($\text{perimeter} = 4.0 * \text{side}$).
- Compute the coloring cost using ($\text{cost} = \text{area} * \text{rate}$).
- Print the calculated area, perimeter and the total cost to two decimal places.
- End the program.

4 – c code :

```
#include <stdio.h>
```

```
int main() {
```

```
    float side, height, rate;
```

```
    float area, perimeter, cost;
```

```
    // reading input values
```

```
    printf("Enter side: ");
```

```
    scanf("%f", &side);
```

```
    printf("Enter height: ");
```

```
    scanf("%f", &height);
```

```
    printf("Enter rate per sq cm: ");
```

```
    scanf("%f", &rate);
```

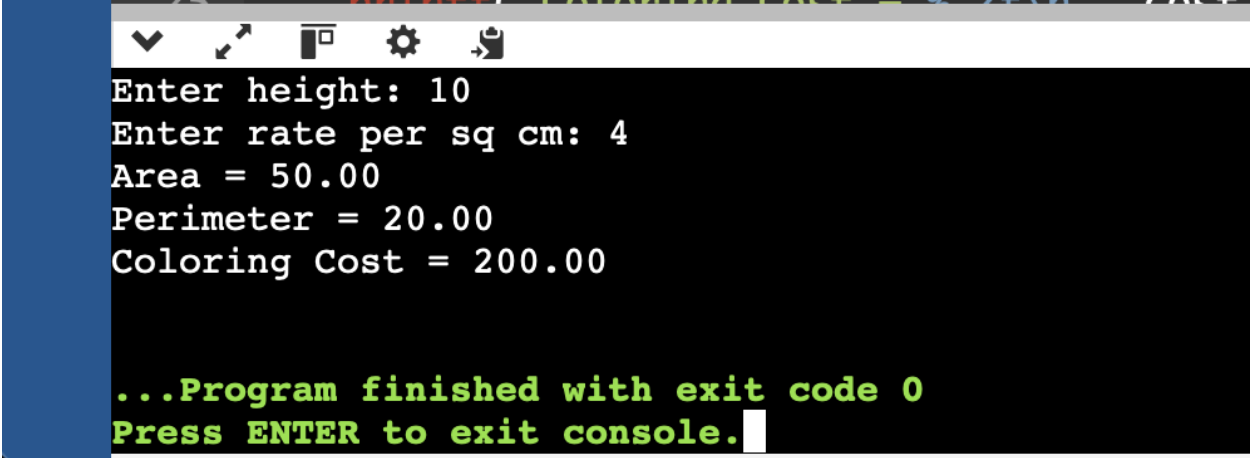
```
    // main calculations
```

```
area = side * height;
perimeter = 4 * side;
cost = area * rate;

// output results
printf("Area = %.2f\n", area);
printf("Perimeter = %.2f\n", perimeter);
printf("Coloring Cost = %.2f\n", cost);

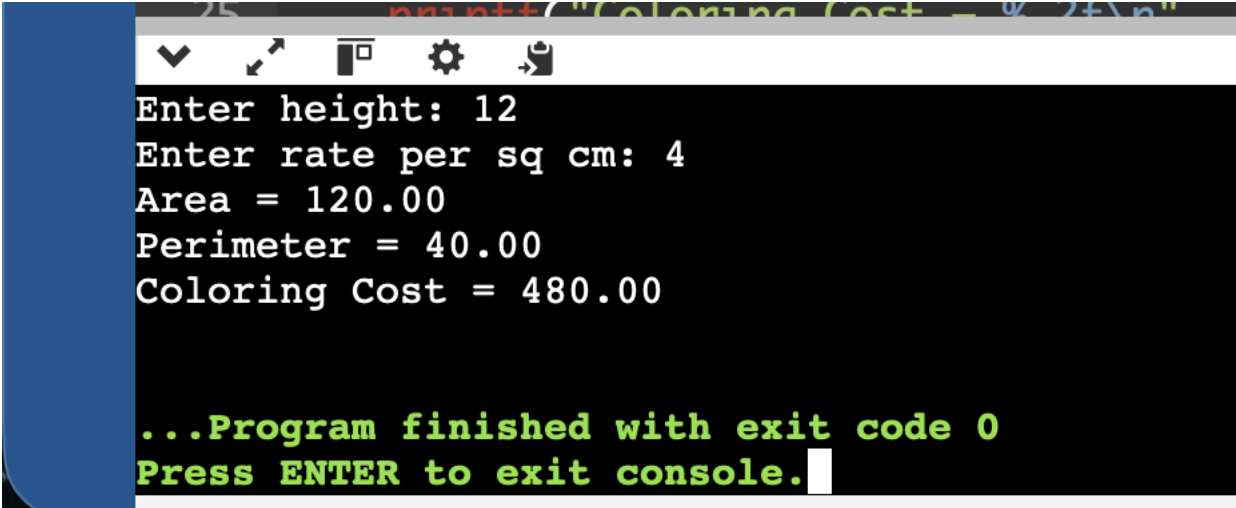
return 0;
}
```

5 – execution results :



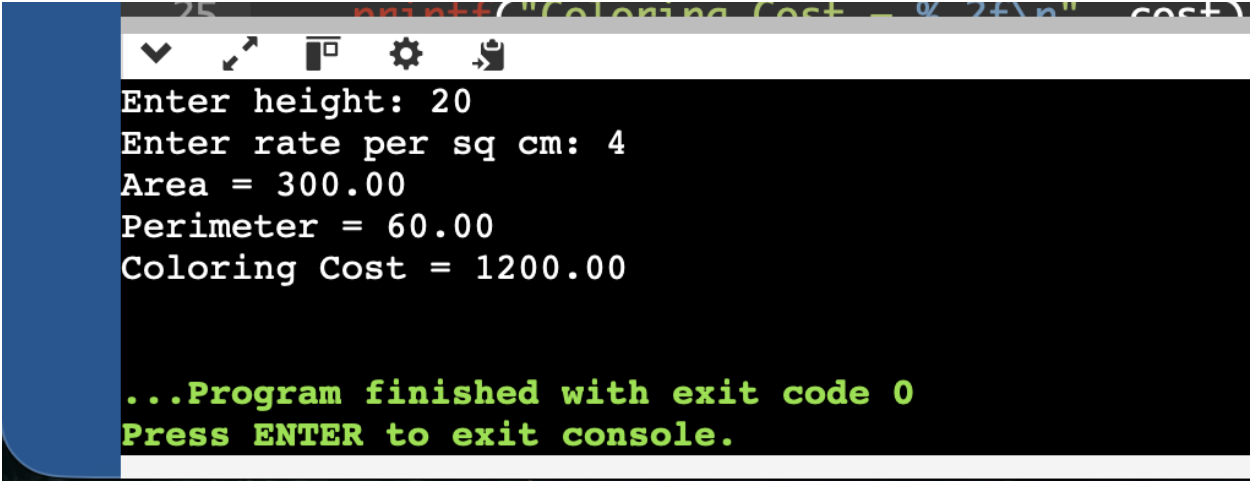
Enter height: 10
Enter rate per sq cm: 4
Area = 50.00
Perimeter = 20.00
Coloring Cost = 200.00

...Program finished with exit code 0
Press ENTER to exit console.



Enter height: 12
Enter rate per sq cm: 4
Area = 120.00
Perimeter = 40.00
Coloring Cost = 480.00

...Program finished with exit code 0
Press ENTER to exit console.



Enter height: 20
Enter rate per sq cm: 4
Area = 300.00
Perimeter = 60.00
Coloring Cost = 1200.00

...Program finished with exit code 0
Press ENTER to exit console.